5. **(a)** Fig. 5.1 shows the variation of the height of a water wave against its horizontal distance at 12 noon.

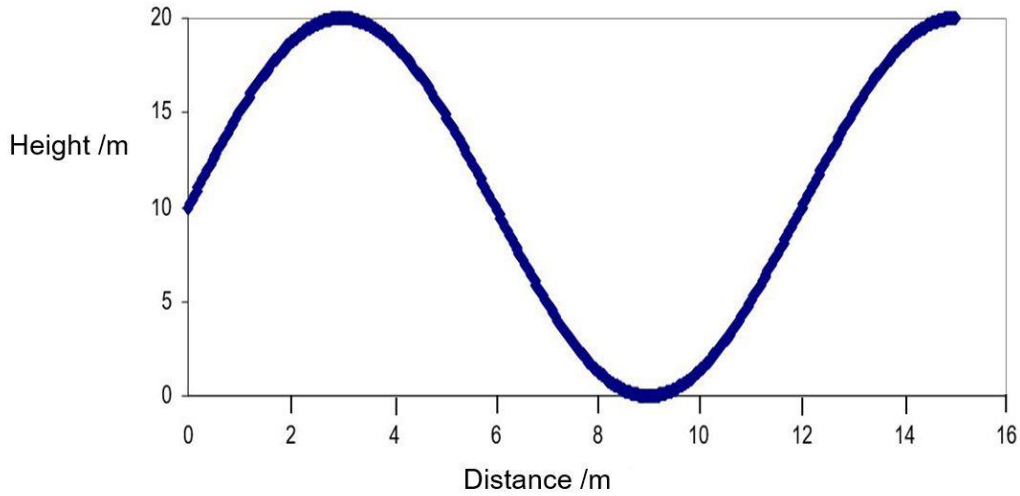

Fig. 5.1

(i) The waves are traveling at the speed of 1.25 m s^{-1}. Show that the frequency of the waves is 0.104 Hz.

[1]

(ii) Two buoys, fixed in position, are floating on the open sea.

The waves take 2.00 minutes to travel directly from one buoy to the other buoy. Determine the phase difference of the two buoys.

phase difference = ... rad [2]

(b) In Fig. 5.2, a narrow beam of monochromatic light of wavelength 590 nm passes through two different diffraction gratings A and B positioned perpendicular to each other. Assume that the distance between the two gratings is negligible.

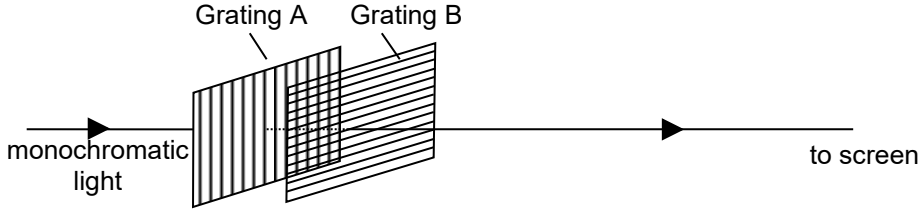

Fig. 5.2

As a result of this arrangement, a diffraction pattern is formed on a screen placed 2.5 m away from the gratings. The pattern is shown in Fig. 5.3.

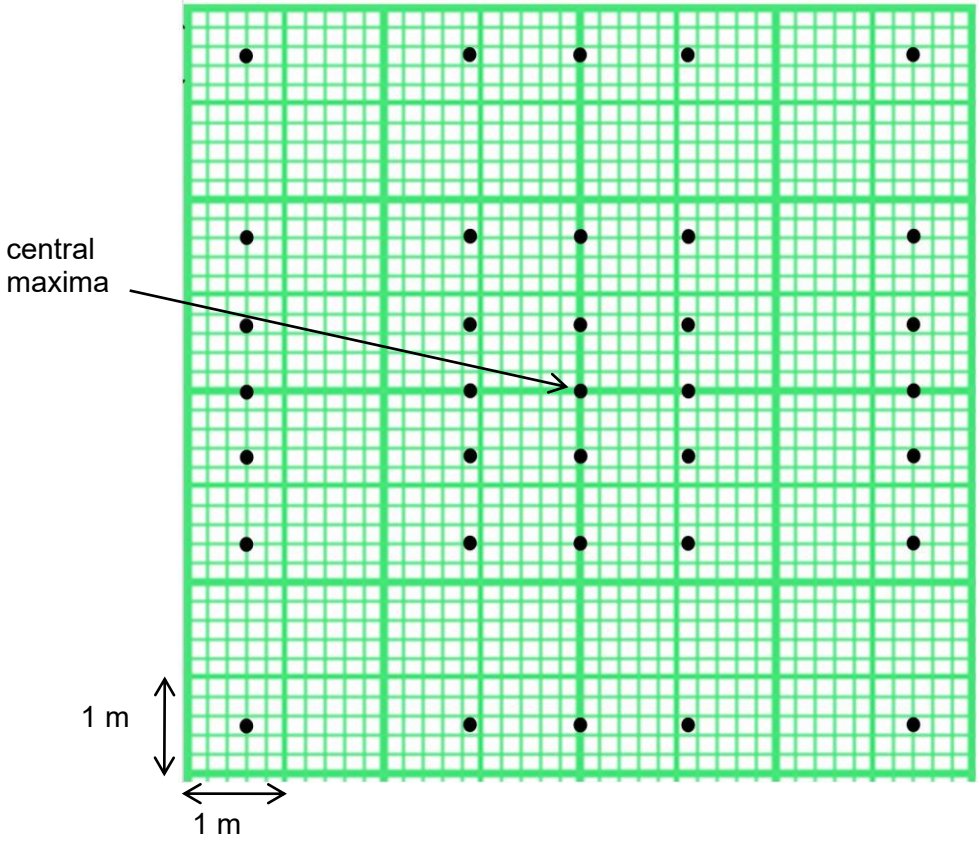

Fig. 5.3

(i) Explain the meaning of the term *diffraction*.

.....

[1]

.....

(ii) By taking appropriate measurements from Fig. 5.3, determine the number of lines per unit length on diffraction grating B.

number = ... lines m^{-1} [3]

(iii) Based on the diffraction pattern shown in Fig. 5.3, explain qualitatively what can be deduced about the slit separation of both diffraction gratings.

.....

.....

..... [2]

(iv) The two diffraction gratings are now replaced with a fine nylon mesh with 24000 nylon threads per metre of the mesh. Suggest one way in which the diffraction pattern observed would be different.

.....

..... [1]